



VIT

Vellore Institute of Technology

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
FALL SEMESTER 2026-2027

REG.NO.: 26MA10076

SLOT: E2+TE2

Programme Name & Branch : MTECH - Computer Science and Engineering(AI&ML)
Course Code and Course Name : MACSE519 & Machine Learning and Applications
Faculty Name(s) : Prof. Arunkumar T, Prof. Sanjiban Sekar Roy
Class Number(s) : VL2026270106-064
Date of Examination : 13th August 2026
Exam Duration : 90 minutes
Maximum Marks: 50

General instruction(s):

- Answer All Questions(10XS=50)
- M - Max mark; CO - Course Outcome; BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
- Course Outcomes: (Type the CO statements covered in this question paper. Use the CO number as per the syllabus copy)

Q. No	Question	M	CO	BL
1.	The Netflix Prize was an open competition for the best collaborative filtering algorithm to predict user ratings for films. Competitors were given an $n \times d$ ratings matrix R; entry R_{jk} is user j's rating of movie k. Because users only watch a small fraction of the movies, most entries in R are unobserved, hence filled with a default value of zero. Latent factor analysis attempts to predict missing ratings by replacing R with a low-rank approximation. Design and develop a decision tree approach to classify the user's movie rating metric, along with number of times a movie is watched to suggest Movie Recommendation approach. Use appropriate dataset and explain this case study.	10	1	4
2.	Given the sample points below, draw and label two lines: the decision boundary learned by a hard-margin SVM and the decision boundary learned by a soft-margin SVM. Define the qualitative difference between hard- and soft-margin SVMs.) Label the two lines clearly. Explain working model of SVM using patient's body temperature, high blood pressure to classify patient's illness as normal or abnormal	10	1	6



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3.	Consider the training dataset given in the following table. Use Weighted k-NN and determine the class. What are the Test instances and explain.	10	2	6																																													
<table><tr><th>S.No.</th><th>CGPA</th><th>Assessment</th><th>Project Submitted</th><th>Result</th></tr><tr><td>1</td><td>9.2</td><td>85</td><td>8</td><td>Pass</td></tr><tr><td>2</td><td>8</td><td>80</td><td>7</td><td>Pass</td></tr><tr><td>3</td><td>8.5</td><td>81</td><td>8</td><td>Pass</td></tr><tr><td>4</td><td>6</td><td>45</td><td>5</td><td>Fail</td></tr><tr><td>5</td><td>6.5</td><td>50</td><td>4</td><td>Fail</td></tr><tr><td>6</td><td>8.2</td><td>72</td><td>7</td><td>Pass</td></tr><tr><td>7</td><td>5.8</td><td>38</td><td>5</td><td>Fail</td></tr><tr><td>8</td><td>8.9</td><td>91</td><td>9</td><td>Pass</td></tr></table>					S.No.	CGPA	Assessment	Project Submitted	Result	1	9.2	85	8	Pass	2	8	80	7	Pass	3	8.5	81	8	Pass	4	6	45	5	Fail	5	6.5	50	4	Fail	6	8.2	72	7	Pass	7	5.8	38	5	Fail	8	8.9	91	9	Pass
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4.	(a) Explain in detail about Support Vector Machines with algorithms and state the role of hyper planes. (b) Define simple linear regression using a graph explaining slope and intercept also Explain rise, run, and slop in graph	10 (5+5)	2	5																																													
5.	Apply Candidate Elimination Hypothesis for the following dataset. <table><tr><th>Sky</th><th>AirTemp</th><th>Humidity</th><th>Wind</th><th>Water</th><th>Forecast</th><th>EnjoySport</th></tr><tr><td>Sunny</td><td>Warm</td><td>Normal</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>Rain</td><td>Cold</td><td>High</td><td>Strong</td><td>Warm</td><td>Change</td><td>No</td></tr><tr><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Cool</td><td>Change</td><td>Yes</td></tr></table> Explain step by step on working of this model using Learned Version Space	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport	Sunny	Warm	Normal	Strong	Warm	Same	Yes	Sunny	Warm	High	Strong	Warm	Same	Yes	Rain	Cold	High	Strong	Warm	Change	No	Sunny	Warm	High	Strong	Cool	Change	Yes	10	2	5										
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